

Predicting the Most Likely Scorer in the 2026 World Cup Final

A leakage-free player match analysis of Spain vs Argentina using FIFA Training Centre post-match data

Executive finding

Before the final, the regularised logistic model ranked **Lionel MESSI (Argentina)** as the most likely scorer at **69.5%** over 90 minutes. The 95% match bootstrap interval was 21.7% to 96.2%. The actual scorer, **Ferran TORRES**, ranked 12 before kickoff. He scored in extra time as Spain won 1-0, illustrating why a probability ranking can be useful without being certain.

Analytic question

Using only information available when the starting lineups became known, which player in the Spain vs Argentina final had the highest probability of scoring at least once? The unit of analysis is a player selected in a matchday squad, including unused substitutes. This makes the task harder and more realistic than ranking players who were already known to appear.

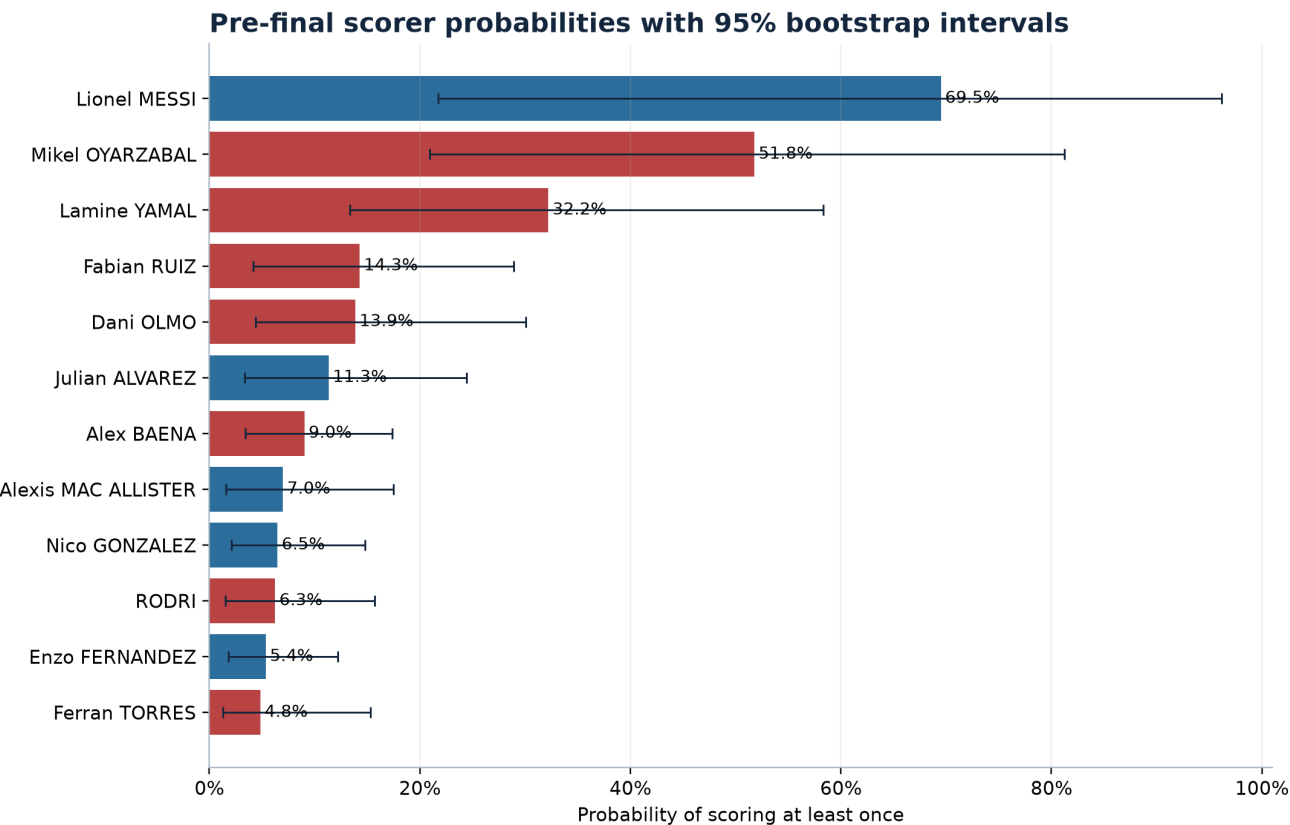


Figure 1. Probabilities refer to at least one goal in 90 minutes. Intervals resample whole matches.

Data acquisition and validation

The pipeline uses six relational CSV tables from the open Worldcup26 repository, pinned to commit 7cc05cdc046f. The source converts all 104 FIFA Training Centre reports into structured match, team, player, lineup and attempt data. The analysis includes 5,392 player matchday records and 2,571 attempts at goal. The supplied raw snapshot lets the project run offline, while --refresh-data repeats acquisition from the pinned URLs and writes SHA-256 hashes.

Validation checks

Check	Result	Evidence
104 tournament matches	PASS	rows=104
Unique match identifiers	PASS	unique_ids=104, unique_numbers=104
Two team rows per match	PASS	range=2..2
Complete reported matchday squads	PASS	team_squad_range=24..26; 195 of 208 contain 26 players
Unique appearance identifiers	PASS	unique=5392, rows=5392
Appearance match keys resolve	PASS	unmatched=0
Attempt player and match keys resolve	PASS	unmatched_appearances=0
Player roster keys resolve	PASS	unmatched=0
Required team metrics present	PASS	found=['Attempts at Goal (On Target)', 'Goals', 'xG (Expected Goals)']
Final identified as Spain vs Argentina	PASS	Spain vs Argentina
Final withheld from feature histories	PASS	Feature construction updates histories only after each prediction row is created.

Leakage control

For every match, the pipeline calculates player and team histories before adding that match's outcomes. The final is match 104 and never contributes to training features. Walk-forward testing trains on earlier matches and predicts the next knockout match in chronological order. This design avoids the future information leakage caused by random row splitting.

Creative and systematic variables

The 19 predictors represent opportunity, individual threat, recent form and match context. Opportunity combines the announced starter status with a player's past selection, appearance and starting rates to estimate minutes. Threat uses smoothed goals, attempts and shots on target per 90 minutes, including separate three match form. Context adds the team's prior goals, expected goals and attempts, the opponent's prior goals, expected goals and shots on target allowed, rest days, position and knockout status. Bayesian style pseudo counts stabilise players with few minutes.

Model comparison and selection

Three viable approaches were tested on matches 73 to 103. The historical rate baseline is transparent but ignores team and opponent context. The random forest captures non-linear interactions but may overfit rare scorer events. It achieved a narrowly lower Brier score, while the regularised logistic model achieved better log loss, ROC AUC and top-ranked scorer accuracy. The logistic model was selected because the Brier difference was very small and its probability effects are easier to explain and audit.

Model	Brier	Log loss	ROC AUC	Top 1 hit	Top 3 hit
Random forest	0.0322	0.1244	0.864	46.4%	78.6%
Regularised logistic	0.0323	0.1232	0.877	50.0%	82.1%
Historical rate baseline	0.0331	0.1260	0.854	46.4%	71.4%

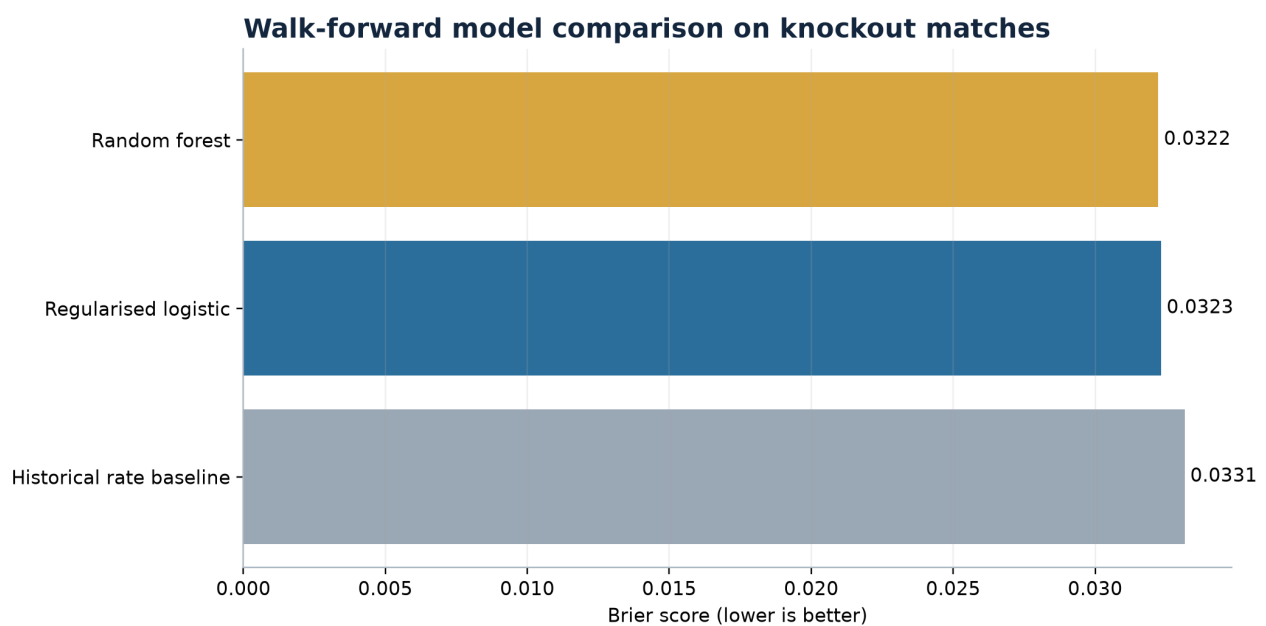


Figure 2. Lower Brier scores indicate more accurate probability forecasts.

Assumption and diagnostic checks

The scorer outcome is rare: 4.2% of pre-final squad rows. L2 regularisation and smoothed rate features reduce unstable estimates. The largest approximate variance inflation factor was 47.3 for expected minutes, so correlated rate variables remain a limitation even with regularisation. The calibration plot checks whether predicted probabilities match observed frequencies. Player rows within the same match are not independent, so uncertainty uses a cluster bootstrap that resamples complete matches rather than individual players.

Interpretation of the final forecast

Rank	Player	Team	Role	90 min	120 min	95% interval
1	Lionel MESSI	Argentina	Starter	69.5%	79.5%	21.7%-96.2%
2	Mikel OYARZABAL	Spain	Starter	51.8%	62.2%	20.9%-81.2%
3	Lamine YAMAL	Spain	Starter	32.2%	40.4%	13.3%-58.4%
4	Fabian RUIZ	Spain	Starter	14.3%	18.6%	4.2%-28.9%
5	Dani OLMO	Spain	Starter	13.9%	18.0%	4.4%-30.1%
6	Julian ALVAREZ	Argentina	Starter	11.3%	14.8%	3.4%-24.5%
7	Alex BAENA	Spain	Starter	9.0%	11.9%	3.4%-17.4%
8	Alexis MAC ALLISTER	Argentina	Starter	7.0%	9.2%	1.6%-17.5%
9	Nico GONZALEZ	Argentina	Starter	6.5%	8.5%	2.1%-14.8%
10	RODRI	Spain	Starter	6.3%	8.2%	1.5%-15.7%
11	Enzo FERNANDEZ	Argentina	Starter	5.4%	7.1%	1.8%-12.2%
12	Ferran TORRES	Spain	Bench	4.8%	6.4%	1.3%-15.3%

What the result means

The highest probability is still modest because a football goal is a rare event shared among many players. Lionel MESSI's ranking reflects both scoring threat and expected playing time. The 120 minute sensitivity raises every player's chance but does not assume that a starter necessarily remains on the field for extra time. The probabilities should support comparison, not be read as guarantees or betting advice.

Post-match audit

Spain defeated Argentina 1-0 after extra time. Ferran TORRES scored the only goal. The model placed him at rank 12, so the single realised match did not match the top ranked player. This does not by itself invalidate a probability model; model quality is judged across the earlier walk-forward sample. The miss shows the importance of substitute uncertainty and low-frequency events in football.

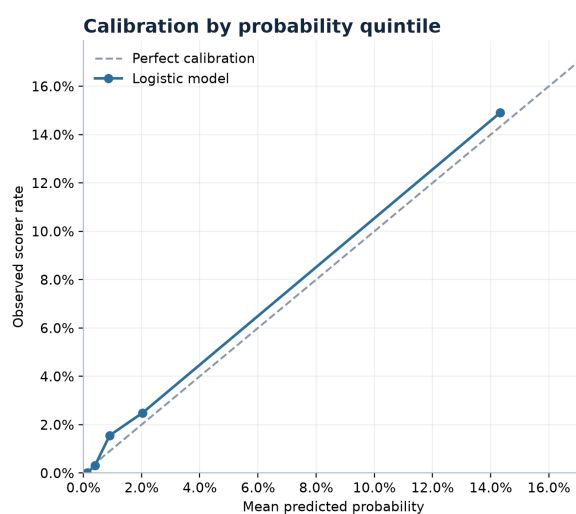


Figure 3. Observed scorer frequency by model probability quintile.

Limitations and viable extensions

The model uses one tournament, so each finalist contributes only six pre-final matches. It knows the announced starting lineup but estimates substitute use from tournament history. It lacks player level expected goals, penalty taker status, injuries, tactical matchups and club season form. The public reports may contain extraction errors; for example, event minutes can differ by one minute across providers. Extra time sensitivity assumes a proportional hazard and cannot predict substitutions or red cards. A stronger future study would merge club shot quality, bookmaker team goal expectations and confirmed injury data, then test the model on multiple tournaments.

Reproducibility

Install the packages in requirements.txt and run `python run_analysis.py`. The command validates the raw tables, rebuilds every feature in chronological order, compares models, writes the prediction tables and diagnostics, regenerates all figures, and creates this PDF. Run `python run_analysis.py --refresh-data` to reacquire the pinned CSV snapshot when internet access is available.

References

- Alamy. (2026). Worldcup26: Open FIFA World Cup 2026 data derived from FIFA Training Centre post-match reports. GitHub. <https://github.com/Alamy/Worldcup26>
- FIFA Training Centre. (2026). Post-match summary report: Spain vs Argentina, match 104. <https://www.fifatrainingcentre.com/media/native/tournaments/fifa-world-cup/2026/PMSR-M104-ESP-V-ARG.pdf>
- FIFA. (2026). Spain crowned champions as curtain falls on FIFA World Cup 2026. <https://inside.fifa.com/organisation/news/new-york-jersey-stadium-spain-world-champions-mbappe-haaland>